

CENG381 Stochastic Processes

CTMC and Birth-Death Processes — Practice 1 (Answer Key)

Q1 Solution

a) Rate matrix Q:

	0	1	2
0	[-2	2	0]
1	[3	-4	1]
2	[0	2	-2]

Row sums: 0: $-2+2+0 = 0$ ✓
1: $3-4+1 = 0$ ✓
2: $0+2-2 = 0$ ✓

b) Stationary distribution ($\pi \cdot Q = 0$):

Column equations of $\pi \cdot Q = 0$:

$$j=0: -2\pi_0 + 3\pi_1 = 0 \Rightarrow \pi_1 = (2/3)\pi_0 \quad \dots(I)$$

$$j=2: \pi_1 - 2\pi_2 = 0 \Rightarrow \pi_2 = \pi_1/2 \quad \dots(II)$$

$$\Rightarrow \pi_2 = (1/3)\pi_0$$

$$\text{Normalise: } \pi_0 + (2/3)\pi_0 + (1/3)\pi_0 = 2\pi_0 = 1$$

$$\pi_0 = 1/2, \quad \pi_1 = 1/3, \quad \pi_2 = 1/6$$

c) Interpretation and expected return time:

Idle: 50% of the time. Working: 33.3% of the time. Broken: 16.7% of the time.

Expected return time to Working state = $1/\pi_1 = 3$ hours.

Q2 Solution

a) Rate matrix Q (states 0–3):

	0	1	2	3
0	[-4	4	0	0]
1	[2	-5	3	0]
2	[0	3	-5	2]
3	[0	0	4	-4]

b) Detailed balance and stationary distribution:

$$\lambda_n \cdot \pi_n = \mu_{n+1} \cdot \pi_{n+1}:$$

CENG381 Stochastic Processes

CTMC and Birth-Death Processes — Practice 1 (Answer Key)

```
n=0: lambda_0*pi_0 = mu_1*pi_1  =>  4*pi_0 = 2*pi_1  =>  pi_1
= 2*pi_0
n=1: lambda_1*pi_1 = mu_2*pi_2  =>  3*2*pi_0 = 3*pi_2  =>
pi_2 = 2*pi_0
n=2: lambda_2*pi_2 = mu_3*pi_3  =>  2*2*pi_0 = 4*pi_3  =>
pi_3 = pi_0

Normalise: pi_0 + 2*pi_0 + 2*pi_0 + pi_0 = 6*pi_0 = 1
pi_0 = 1/6,  pi_1 = 1/3,  pi_2 = 1/3,  pi_3 = 1/6
```

c) $E[N]$ and $P(\text{ward full})$:

$$\begin{aligned} E[N] &= 0 \cdot (1/6) + 1 \cdot (1/3) + 2 \cdot (1/3) + 3 \cdot (1/6) \\ &= 0 + 1/3 + 2/3 + 1/2 = 1 + 1/2 = 3/2 \end{aligned}$$

$$P(\text{ward full}) = \pi_3 = 1/6 \approx 16.7\%$$

Note the symmetry: $\pi_0 = \pi_3 = 1/6$, $\pi_1 = \pi_2 = 1/3$. This arises because the birth and death rates are mirror-symmetric around the midpoint (both increase linearly from the boundaries).

Q3 Solution

a) Rate matrix and stationary distribution:

States: 0 = water, 1 = land. $q(0,1) = \beta = 1$ (water $\rightarrow$ land), $q(1,0) = \alpha = 3$ (land $\rightarrow$ water).

$$\begin{array}{cc} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} -1 & 1 \\ 3 & -3 \end{bmatrix} \end{array}$$

$$\pi \cdot Q = 0:$$

$$j=0: -\pi_0 + 3\pi_1 = 0 \Rightarrow \pi_0 = 3\pi_1$$

$$\text{Normalise: } 3\pi_1 + \pi_1 = 1 \Rightarrow \pi_1 = 1/4, \quad \pi_0 = 3/4$$

The frog spends 75% of its time in water and 25% on land.

b) Embedded discrete chain P ($r = 3$):

$$P(x,y) = q(x,y)/r \quad \text{for } x \neq y; \quad P(x,x) = 1 - q(x)/r$$

CENG381 Stochastic Processes

CTMC and Birth-Death Processes — Practice 1 (Answer Key)

$$\begin{aligned}P(0,1) &= q(0,1)/3 = 1/3, & P(0,0) &= 1 - 1/3 = 2/3 \\P(1,0) &= q(1,0)/3 = 3/3 = 1, & P(1,1) &= 0\end{aligned}$$

Stationary of P: solve $\pi \cdot P = \pi$.

$$\begin{aligned}\pi_0 &= (2/3)\pi_0 + 1\pi_1 \Rightarrow (1/3)\pi_0 = \pi_1 \\ \pi_0 + \pi_1 &= 1 \Rightarrow (4/3)\pi_0 = 1 \Rightarrow \pi_0 = 3/4, \quad \pi_1 = 1/4 \quad \checkmark\end{aligned}$$

c) Eigenvalue of Q and time-evolution formula:

$$\begin{aligned}\text{Eigenvalues of Q: } \det(Q - \lambda I) &= 0 \\ (-1-\lambda)(-3-\lambda) - 3 &= 0 \\ \lambda^2 + 4\lambda + 3 - 3 &= 0 \\ \lambda(\lambda + 4) &= 0 \\ \lambda_1 &= 0, \quad \lambda_2 = -4 \\ P_1(t) &= \pi_1 + C e^{-4t} \\ P_1(0) = 1 &= \pi_1 + C = 1/4 + C \Rightarrow C = 3/4 \\ P_1(t) &= 1/4 + (3/4)e^{-4t}\end{aligned}$$

Time for $|P_1(t) - \pi_1| < 0.05$:

$$\begin{aligned}(3/4)e^{-4t} &< 0.05 \\ e^{-4t} &< 0.05 \cdot (4/3) = 1/15 \\ -4t &< \ln(1/15) = -\ln(15) \\ t &> \ln(15)/4 = 2.708/4 = 0.677 \text{ hours} \quad \approx 40.6 \text{ minutes}\end{aligned}$$